Sami Bekuma

Silver Spring, MD • samibekuma.tmc@gmail.com • 678.793.4508 • linkedin.com/in/samibekuma •https://samibekuma.vercel.app/

Professional Summary

Machine Learning Engineer with experience building and deploying production ML and LLM systems serving more than 20 million daily user interactions. Specialized in NLP, RAG pipelines, and intent classification, with hands-on expertise in cost optimization using Azure OpenAI, AWS Bedrock and open-source models. Focused on improving model reliability and accelerating deployment cycles for high-traffic platforms through automated evaluation, efficient routing strategies, and MLOps best practices.

PROFESSIONAL EXPERIENCE

Machine learning Engineer (Contractor)

Comcast - Washington, DC 2024 –present

- **Designed and built an end-to-end ML/NLP validation ecosystem** for the Xfinity Voice Remote AI platform, supporting **20M+ daily voice queries** across North America and Europe, ensuring high reliability and model performance.
- **Architected and developed a scalable automated evaluation framework (Python)** integrated with **Azure CI/CD**, leveraging **LLMs to evaluate model outputs** (factuality, hallucination, relevance, toxicity), **reducing validation cycles by 85%** and becoming the team's primary AI validation platform.
- **Built and maintained robust test automation frameworks** for NLP services using **Java + Spring Boot**, enabling **functional, regression, and API-level testing** for newly developed features.
- **Developed an NLP Integrity Validator service** to detect anomalies in logical forms generated by NLP and prediction systems, proactively preventing downstream failures and improving end-user interaction quality.
- Designed and built production-driven validation pipelines leveraging live queries from logs and Slack streams, eliminating dependence on labeled datasets and enabling continuous real-world performance evaluation of NLP systems.
- **Built CI/CD automation pipelines (Jenkins, Azure DevOps)** to execute nightly test runs and automatically distribute **test reports to stakeholders**, improving visibility and release confidence.
- **Deployed and tested ML models and microservices in Kubernetes**, configuring multiple environments and parameters to support **A/B testing and experimentation**.
- **Engineered a full-stack test reporting platform (Angular + MongoDB)** to deliver **interactive, user-friendly test insights** for engineers and product managers.
- **Collaborated cross-functionally** with product managers, ML engineers, and backend teams to improve **test coverage, automation strategies, and overall system quality**.
- **Annotated and structured evaluation datasets using internal tools (NAT)**, ensuring high-quality test cases with correct schemas and expected outputs.

Lead-tutor

Tutoring Club of McLean, *McLean, VA* 2022 – 2024

- Mentored students in Java and Python programming, focusing on core programming concepts and problem-solving skills.
- Developed structured lesson plans and supported SAT/ACT preparation, improving students' technical confidence and exam readiness.

Assistant Manager

Reef Parking Technology, *Washington, DC* 2020 – 2022

- Led and managed a cross-functional team of 15+ parking attendants, valets, and maintenance staff, providing training, mentorship, and optimizing daily operations to ensure efficient service delivery.
- Automated routine operational tasks to reduce manual workload and increase accuracy to satisfy customer needs.

SKILLS

Languages & Frameworks:
Python (Advanced), JavaScript, React, Java, SQL, Spring Boot
Data & Machine Learning:
Python (Pandas, NumPy, Scikit-learn), Transformers (Hugging Face), LangChain, LlamaIndex, SentenceTransformers, SpaCy, NLTK, MLflow, VLLM, KServe, RAG Systems, Model Evaluation & Benchmarking, LLM-as-a-Judge
Analytics & Dashboarding:
Dashboard Development, Data Visualization, Grafana, Business Metrics Analysis, Experimentation & KPI Tracking
DevOps & Cloud:
AWS (SageMaker, EC2, S3, Lambda), Azure, Docker, Kubernetes, Git, Jenkins, GitHub Actions
Soft Skills:
Problem Solving, Business Acumen, Cross-functional Collaboration, Communication, Data Quality Mindset

TECHNICAL PROJECTS

AI-Powered Doctor Appointment Booking System (React, Node.js, Express, MySQL, Google Gemini)

- Designed and implemented a **production-style healthcare scheduling platform** that replaces manual and phone-based appointment booking with a structured, self-service workflow.
- Integrated an **AI-powered chatbot (Google Gemini)** that interacts with backend APIs and the database to **create and cancel appointments in real time**, in addition to navigation and user assistance.
- Designed **backend scheduling and conflict-prevention logic** using Node.js, Express, and relational data modeling in MySQL.

Automated Real-Time Data Ingestion Pipeline with AWS and OpenSearch

- Designed and deployed an end-to-end cloud data ingestion pipeline using **AWS** services (**S3**, **SQS**, **Lambda**, **OpenSearch Service**) to automate real-time data processing and analytics.
- Configured an **event-driven architecture** to trigger processing on new **S3** uploads, enabling immediate parsing and indexing into **OpenSearch** for scalable search and visualization.

EDUCATION

GEORGE MASON UNIVERSITY, The Volgenau School of Engineering, Fairfax, VA Dec 2023
BS in Computer Science Grade: 3.64
Honors & Awards: 2022 All-Virginia Academic Team Award, Dean's List, Presidential Scholar

CERTIFICATIONS

Meta Front-End and Back-End Developer Professional Certificate – Coursera | 2023

Relevant Coursework: HTML, CSS, JavaScript, React, Version Control, Frontend & Backend Fundamentals

- HTML and CSS in Depth
- Version Control
- React Basics
- Introduction to Back-End Development
- Programming with JavaScript
- Introduction to Front-End Development